

Business Problem

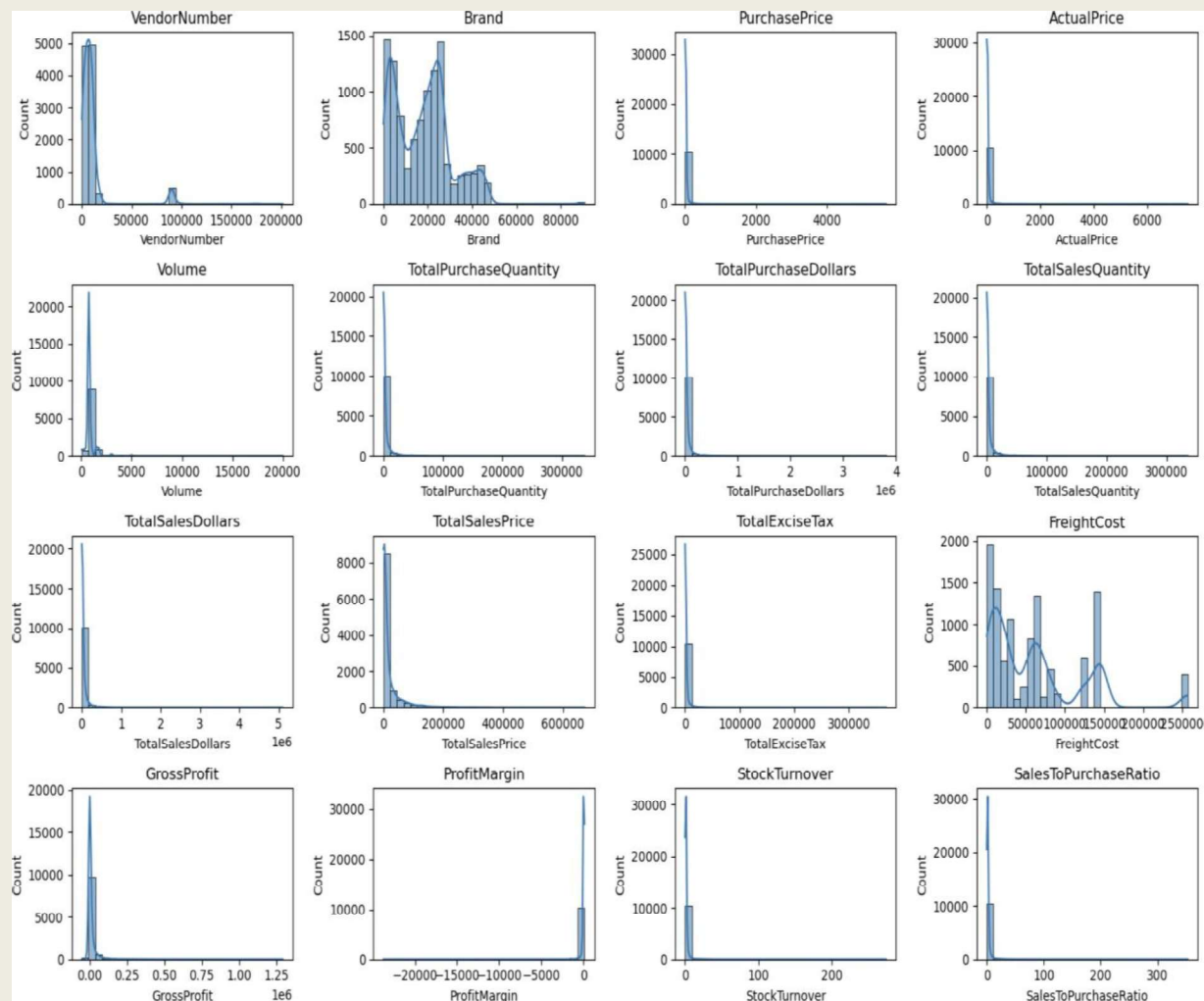
Efficient inventory and sales management play a vital role in maximizing profitability within the retail and wholesale sector. Businesses must minimize losses arising from ineffective pricing strategies, slow inventory movement, and excessive reliance on specific vendors. This analysis aims to:

- Identify underperforming brands that may benefit from pricing revisions or promotional strategies.
- Determine the top vendors driving overall sales and gross profit.
- Evaluate how bulk purchasing influences per-unit procurement costs.
- Analyze inventory turnover to reduce carrying costs and enhance operational efficiency.
- Examine profitability differences between high-performing and low-performing vendors.

Exploratory Data Analysis Insights

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02



Negative and Zero Values

- **Gross Profit:** The minimum gross profit recorded is **-52,002.78**, indicating that certain products generated losses. This may result from high procurement costs, excessive discounts, or selling products below their purchase price.
- **Profit Margin:** The minimum profit margin is $-\infty$, suggesting cases where sales revenue was zero or insufficient to cover total costs, resulting in extremely negative margins.
- **Sales Quantity and Sales Revenue:** Some products recorded **zero sales quantity and zero sales value**, implying that inventory was purchased but never sold. These items could represent slow-moving or obsolete stock, contributing to inefficient inventory management.

Outliers Identified Through High Standard Deviations

- **Purchase and Selling Prices:** The maximum purchase and actual selling prices (**5,681.81** and **7,499.99**, respectively) are substantially higher than their average values (**24.39** and **35.64**). This indicates the presence of premium or high-value products within the portfolio.
- **Freight Cost:** Freight expenses vary significantly, ranging from **0.09**

to **257,032.07**. Such a wide spread may be attributed to differences in shipment sizes, bulk purchases, transportation inefficiencies, or inconsistent logistics costs.

- **Stock Turnover:** Inventory turnover values range from **0 to 274.5**, highlighting considerable differences in product demand. While some products are sold quickly, others remain in stock for extended periods. A turnover ratio greater than **1** indicates that sales exceeded current purchase quantities, likely due to fulfilling demand using previously accumulated inventory.

Data Filtering

To improve the accuracy and reliability of the analysis, records containing inconsistent or non-meaningful values were excluded based on the following criteria:

- **Gross Profit ≤ 0 :** Removed transactions that resulted in no profit or financial losses.
- **Profit Margin ≤ 0 :** Excluded records with non-positive profit margins to focus the analysis on profitable transactions.
- **Total Sales Quantity = 0:** Eliminated products with no sales activity, as they do not contribute to revenue generation and may distort performance insights.